SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

I M. MUKKARAM (192524357) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
M. MUKKARAM

Annexure A

Technical Presentation

1. **Reactive Agent-Based Traffic Control System**

   Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. **Goal-Based Agent for Healthcare Diagnosis**

   Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. **Utility-Based Agent for E-Commerce Recommendations**

   Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. **Learning Agent for Personalized Education System**

   Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. **Multi-Agent System for Disaster Management**

   Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness


GPS Map Camera
Google
Chennai, Tamil Nadu, India
162, Poonamallee High Rd, Poonamallee,
Chennai, Tamil Nadu 602105, India
Lat 13.024925° Long 80.016637°
Monday, 31/08/2026 02:45 PM GMT +05:30

Title

Designing a Reactive Agent for Smart Traffic Control System

- Artificial Intelligence – Intelligent Agents
- Real-time traffic management using streaming data

Problem Statement

- Traditional traffic signals use fixed timing.
- Traffic density changes continuously throughout the day.
- Fixed signals can cause unnecessary waiting and congestion.
- A **Reactive Agent** can respond immediately to changes in traffic conditions.
- Goal: **Reduce congestion and waiting time by dynamically controlling traffic signals.**

What is a Reactive Agent?

- A **Reactive Agent** is an intelligent agent that selects an action based on the **current state of the environment**.
- **Basic principle:**
- **Traffic Input → Condition/Rule → Action**
- Example:
- If traffic density is high on Road A → Increase green-light duration for Road A.
- It does not need to maintain a complex model of the entire environment.

System Architecture

Real-TimeTrafficSensors
↓
StreamingDataProcessing
↓
ReactiveAgent
↓
TrafficDecision
↓
TrafficSignalController
↓
Updated Traffic Flow

Inputs:

- Number of vehicles
- Vehicle speed

- Queue length
- Emergency vehicles
- Current signal status

Output:

- Green signal
- Red signal
- Yellow signal
- Adjustment of signal duration

Real-Time Traffic Inputs

The system continuously receives data from:

- CCTV cameras
- IoT traffic sensors
- Inductive loop detectors
- GPS/vehicle data

For example:

Road	Vehicles	Queue	Agent Response
Road A	45	High	Extend green
Road B	12	Low	Reduce green
Road C	30	Medium	Normal timing

The streaming system continuously updates these values.

Reactive Decision Rules

The agent uses predefined **IF–THEN rules**.

Example:

- IF traffic density > threshold → **Extend green time**
- IF traffic density < threshold → **Reduce green time**
- IF emergency vehicle detected → **Give priority**
- IF queue length becomes critical → **Increase priority**
- IF traffic is balanced → **Use normal signal cycle**

This allows the system to respond immediately to changing traffic conditions.

Streaming Data Processing

Traffic data arrives continuously rather than as a single dataset.

Sensors → Data Stream → Processing → Agent → Signal Update

The system can process data in short time windows, for example:

Every 5–10 seconds:

1. Collect current traffic data.
2. Calculate traffic density.
3. Compare with predefined thresholds.
4. Select the appropriate signal action.
5. Update the traffic light.
6. Repeat continuously.

This enables **real-time decision-making**.

Technologies Used

Hardware:

- CCTV cameras
- IoT sensors
- Traffic signal controllers

Software:

- Python
- OpenCV
- Apache Kafka / Spark Streaming
- Machine-learning libraries if required
- Traffic simulation tools such as SUMO

Processing:

- Real-time stream processing
- Rule-based decision engine
- Traffic-density calculation

Advantages and Limitations

Advantages

- Fast response to traffic changes
- Reduces unnecessary waiting
- Simple decision mechanism
- Suitable for real-time environments
- Can prioritize emergency vehicles

Limitations

- Depends on accurate sensor data
- Rule-based system may not handle unusual situations well
- Requires reliable communication
- Poorly selected thresholds can cause unstable signal changes

Conclusion

- A Reactive Agent provides **immediate responses to real-time traffic conditions**.
- Streaming data allows continuous monitoring of vehicle density and queue lengths.
- Rule-based decisions dynamically adjust traffic signals.
- The system can reduce congestion, waiting time, and unnecessary stops.
- For larger smart-city systems, reactive control can be combined with predictive or learning-based methods for improved performance.